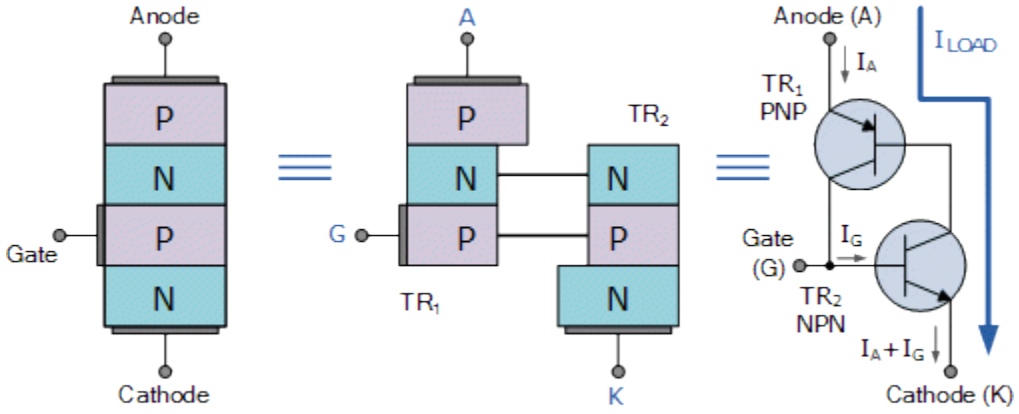
POWER ELECTRONICS

MCQ

FAKHRI JARRAR





إعداد الطالب:

فخري جرار

1) A thyristor (SCR) is a

a) P-N-P device

b) N-P-N device

c) P-N-P-N device

d) P-N device

2)An SCR is a

a) four layer, four junction device

b) four layer, three junction device

c) four layer, two junction device

d) three layer, single junction device

3)Choose the false statement.

a) SCR is a bidirectional device

b) SCR is a controlled device

c) In SCR the gate is the controlling terminal

d) SCR are used for high-power applications

4) The static V-I curve for the SCR is plotted for

a) Ia (anode current) vs Ig (gate current), Va (anode – cathode voltage) as a parameter

b) Ia vs Va with Ig as a parameter

c) Va vs Ig with Ia as a parameter

d) Ig vs Vg with Ia as a parameter

5) If the cathode of an SCR is made positive with respect to the anode & no gate current is applied then

a) all the junctions are reversed biased

b) all the junctions are forward biased

c) only the middle junction is forward biased

d) only the middle junction is reversed biased

6) For an SCR in the reverse blocking mode, (practically)

a) leakage current does not flow

b) leakage current flows from anode to cathode

c) leakage current flows from cathode to anode

d) leakage current flows from gate to anode

7) With the anode positive with respect to the cathode & the gate circuit open, the SCR is said to be in the

a) reverse blocking mode
b) reverse conduction mode
c) forward blocking mode
d) forward conduction mode

8)The forward break over voltage is the

a) anode-cathode voltage at which conduction starts with gate signal applied
b) anode-cathode voltage at which conduction starts with no gate signal applied
c) gate voltage at which conduction starts with no anode-cathode voltage
d) gate voltage at which conduction starts with anode-cathode voltage applied

9)For a forward conducting SCR device, as the forward anode to cathode voltage is increased

a) the device turns on at higher values of gate current
b) the device turns on at lower values of gate current
c) the forward impedance of the device goes on increasing
d) the forward impedance of the device goes on decreasing

10) The forward break over voltage is maximum when

a) Gate current = ∞
b) Gate current = 0
c) Gate current = -∞
d) It is independent of gate current

11) The value of anode current required to maintain the conduction of an SCR even though the gate signal is removed is called as the

a) holding current
b) latching current
c) switching current
d) peak anode current

12) The minimum value of anode current below which it must fall to completely turn-off the device is called as the

a) holding current value
b) latching current value
c) switching current value
d) peak anode current value

13)The latching current is _____ than the holding current

a) lower
b) higher
c) same as
d) negative of

14) The gate characteristics of thyristor is a plot of

a) V_g on the X-axis & I_g on the Y-axis

b) I_g on the X-axis & V_g on the Y-axis

c) V_a on the X-axis & I_g on the Y-axis

d) I_g on the X-axis & V_a on the Y-axis

15) The average output voltage is maximum when SCR is triggered at ωt =

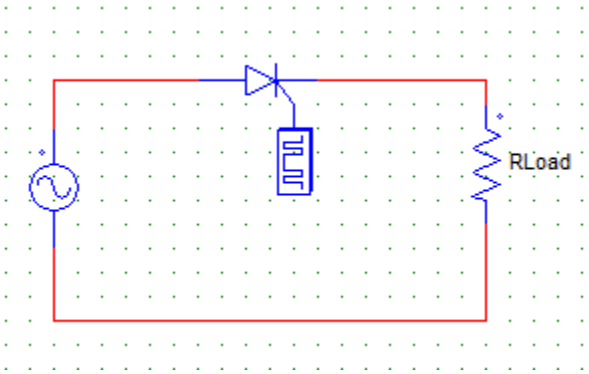
a) π

b) 0

c) π/2

d) π/4

16) Below shown ideal circuit is triggered at a firing angle of α. The voltage across the SCR is zero from ωt =



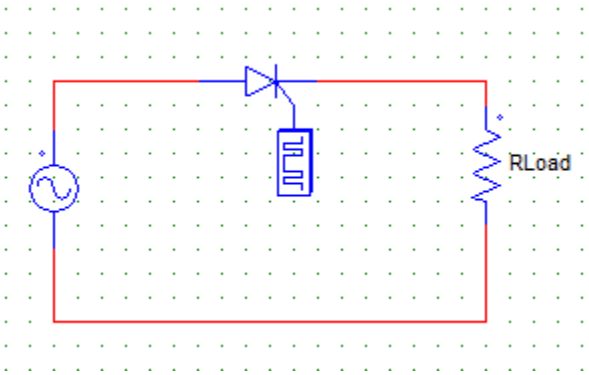
a) 0 to α

b) 0 to π

c) α to 2π

d) α to π

17) Below shown ideal circuit is triggered at a firing angle of α. From ωt = π to 2π+α



a) SCR voltage is zero

b) Load voltage is maximum

c) SCR voltage is –Vmsinα

d) SCR acts like a short circuit

18) A single-phase half-wave thyristor circuit with R load is triggered at an angle of α = 0°. As such, the maximum value of the average output voltage would be given by

Consider Vs = Vm sinωt.

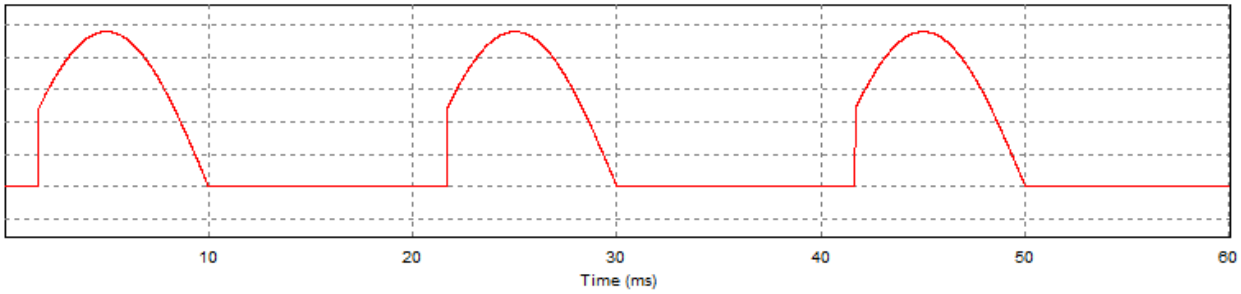
a) Vm

b) 2Vm/π

c) Vm/π

d) Vm/α

19) For a certain SCR configuration, the below shown waveform is obtained. Find the value of the average output voltage with α = 30° & Vs = 240 V.



a) 50.27 V

b) 100.8 V

c) 140 V

d) 120 V

Vm = √2 Vs = 339.41 V

V_{avg} = ∫ Vm Sin ωt d(ωt) where the integration runs from α to π

V_{avg} = Vm/2π (1+cos α).

20) A single phase 230 V, 1 kW heater is connected across a 1-phase 230 V, 50 Hz supply through an SCR. The firing angle of the SCR is adjusted to give a rms voltage of 155 V. Find the power absorbed by the heater element.

a) 250 watts

b) 1 kW

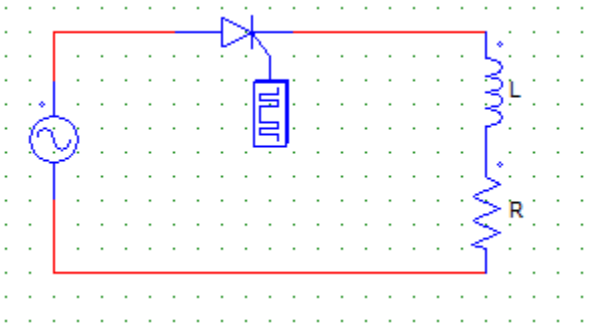
c) 454 watts

d) 378 watts

R (Heater element resistance) = 230^2/1kW = 52.9 Ω

Now, P = $Vrms^2$/R = 155^2/52.9.

21) For the below shown circuit,



The load voltage waveform from ωt = π to ωt = π+α = β (extinction angle) is

a) gradually rising

b) going from positive to negative

c) gradually decaying

d) zero

22) In case of a single-phase half-wave circuit with RL load, with firing angle α and extinction angle β, the thyristor is reversed biased from

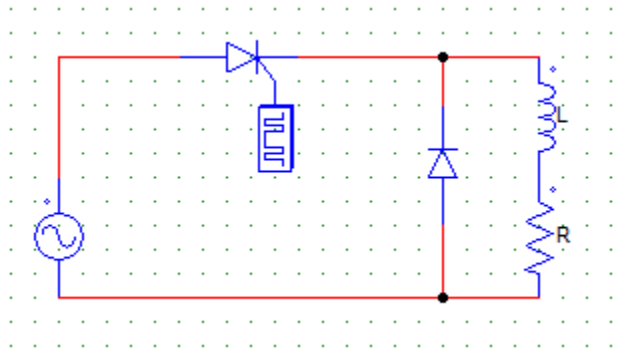
a) β to α

b) β to 2π+α

c) β to 2π

d) β to 2β

23) Choose the correct statement with respect to the below given circuit.



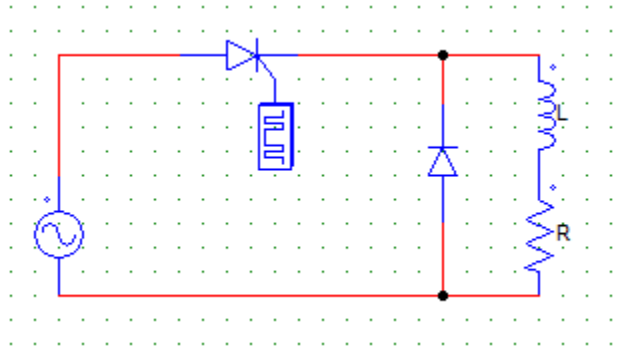
a) The load current can be negative

b) The load voltage can never be negative

c) The load voltage can never be zero

d) The load voltage can never be positive

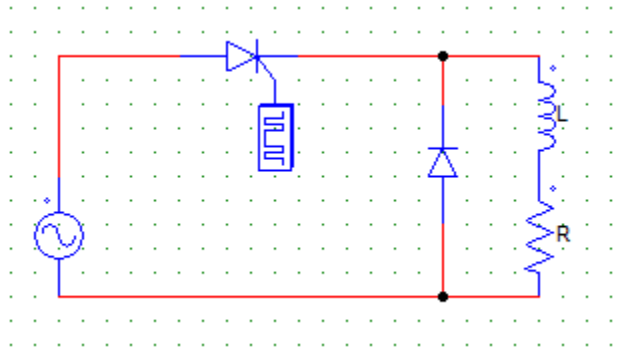
24)In the below given circuit, the FD (Freewheeling diode) is forward biased at ωt =



a) 0
b) α
c) π
d) 2π

SCR has reverse bias voltage and the load current is positive

25) The output voltage waveform of the below given circuit would be the same that obtained from a



a) full-wave R load circuit
b) half-wave R load circuit
c) half-wave RL load circuit
d) full-wave RL load with freewheeling diode

26)In a single-phase half-wave circuit with RL load and a freewheeling diode, the freewheeling period is

a) 0 to π
b) α to π+α
c) π to 2π+α
d) π/2 to 2π-α

27)Choose the incorrect statement with respect to the use of FD in half-wave circuits.

a) Input pf is improved

b) Load current waveform is improved

c) It prevents the load voltage from becoming negative

d) Reduces the reverse voltage (PIV) faced by the SCR

28) A 230V, 50Hz, single-pulse SCR is feeding a RL load with α = 40° and β = 210°. Find the value of average output voltage

a) 54 V

b) 106 V

c) 84 V

d) 32 V

$V_o = V_m/2\pi \times (\cos\alpha - \cos\beta)$

Where $V_m = \sqrt{2}V_s$.

29) A single-pulse transformer with secondary voltage of 230 V, 50 Hz, delivers power to bulb of R = 10 Ω through a half-wave controlled rectifier circuit. For α = 60° and output AC power of 2127 Watts, find the rectification efficiency

a) 98.6 %

b) 42 %

c) 28 %

d) 19 %

$V_o = (V_m/2\pi) \times (1+\cos\alpha) = 77.64$ V

$P_{dc} = V_o^2 \times R = 602.8$ W

Rectification efficiency = $P_{dc}/P_{ac} = 28.32$ %.

30) An SCR rectifier circuit is designed such that the average output voltage is 77.64 V & RMS value of output voltage is 145.873 V. Find the voltage ripple factor.

a) 1.879

b) 0.532

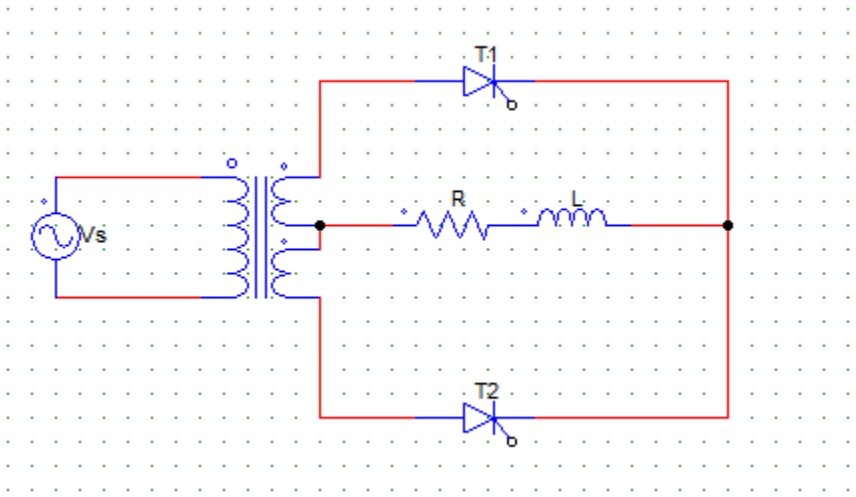
c) 1.590

d) 0.629

FF = 145.873/77.64 = 1.879

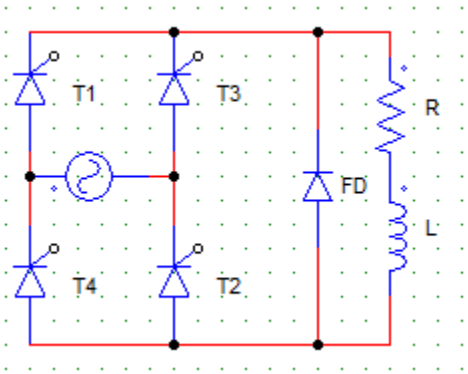
$VRF = \sqrt{(FF^2-1)} = 1.5908$.

31)In the below shown circuit, considering the load to be purely R load, the thyristor T2 would conduct from ωt =



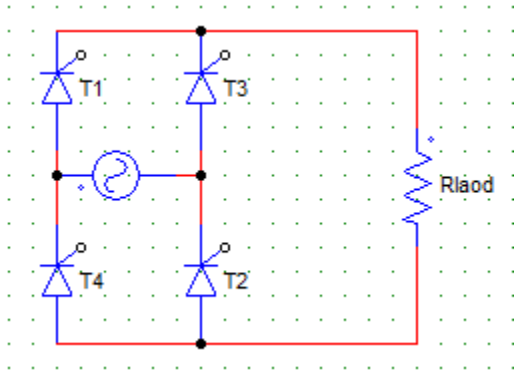
a) 0 to π
b) α to π
c) π+α to 2π
d) π to 2π+α

32) For the below given circuit,



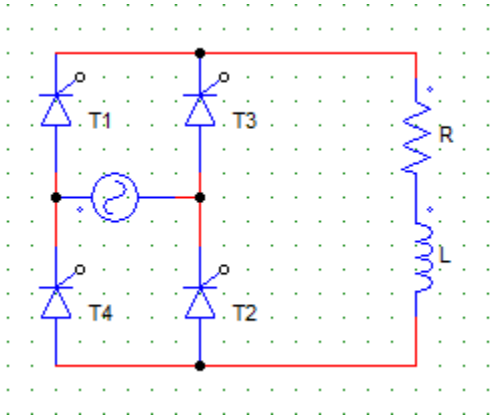
a) T2 and T3 are gated together
b) T1 and T4 are gated together
c) T1 and T3 are gated together
d) T1 and T2 are gated together

33) With respect to the below given circuit, from ωt = 0 to α, ____ thyristor(s) block the forward voltage



a) T1 and T2
b) T2
c) T3 and T4
d) T4

34) The below shown rectifier configuration has continues load current, find the expression for the average value of output voltage when the supply Vs = Vm sinωt is connected.



a) (Vm/π)cosα
b) (2Vm/π)cosα
c) (Vm/π) (1+cosα)
d) (2Vm/π) (1+cosα)

Vo = 1/π x [∫ Vm sinωt d(ωt)] Where the integration runs from α to π+α.

35) A single-phase full converter bridge, is connected to a RLE load. The source has a rms voltage of 230 V and the average load current is 10 A. Find the firing angle for which the power flows from AC source to the DC load. Consider E = 120 V, R = 0.4 Ω.
a) 26.4°
b) 53.2°
c) 142°
d) 39°

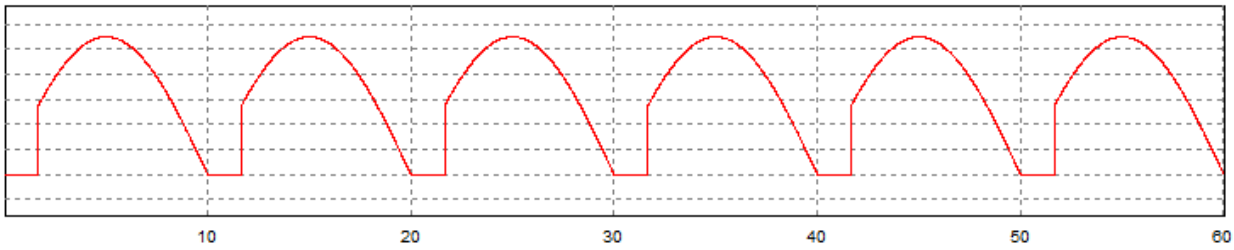
We have,
Vo = E + IR
Vo = 2Vm/π x cosα
Substitute the given values to discover α = 53.208°.

36) A single phase full controlled bridge converter is connected to load having R = 0.4 Ω, E = 120, L = 0.2 mH, Vs = 230V. The RMS value of load current is 10 A for a firing angle of 53.21°. Find the input power factor.
a) 0.5043 lag
b) 0.5391 lag
c) 0.6120 lag
d) Unity

Io = Irms = 10A
Vs x Irms x cosϕ = E Io + $Irms^2$ R.

37) In case of controlled rectifiers, the nature of the load current (continues or discontinuous) depends upon the
a) type of load and firing angle
b) only on the type of load
c) only on the firing angle
d) it is independent of all the parameters

38) A semi-converter circuit gives the following voltage waveform. Find the expression for the average output voltage with Vs = Vm sinωt and firing angle α



a) (Vm/π) cosα
b) (Vm/π) (1+cosα)
c) (2Vm/π) cosα
d) (2Vm/π) (1+cosα)

39) A three-phase, three-pulse, M-3 type controlled converter uses _____ number of SCRs.
a) 1
b) 2
c) 3
d) 4

40) A three-phase, three-pulse, M-3 type controlled converter has firing angle for one of the SCRs set as 15°. This SCR would start conducting at

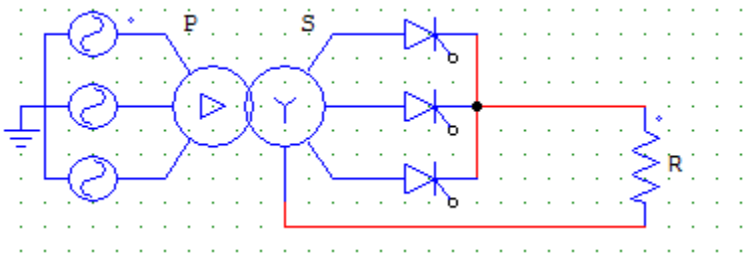
a) 0°
b) 15°
c) 30°
d) 45°

In a three phase controller, the actually conduction starts at 30° + α. Hence, ωt = 30+15 = 45°.

41) A three-phase three pulse type controlled converter is constructed using 3 SCR devices. The circuit is supplying an R load with α < 30°. As such, each SCR device would conduct for

a) 60° each cycle
b) 120° each cycle
c) 180° each cycle
d) 360° each cycle

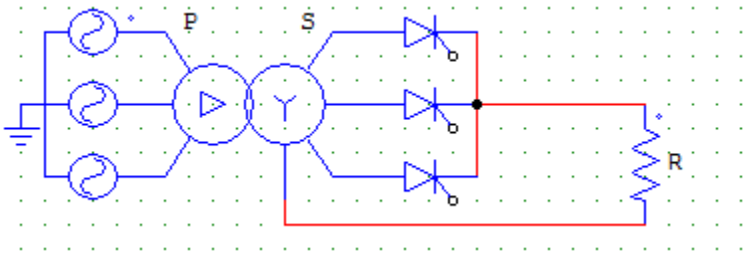
42) Find the expression for average output voltage at R for the below given configuration. Take firing angle as α = 15°, transformer ratio as 1:1:1 and Vmp as the maximum value of phase voltage at the supply.



a) (3Vmp/2π) x cosα
b) (3√3Vmp/2π) x cosα
c) (3√3Vmp/2π) x sinα
d) (3Vmp/2π) x sinα

The circuit is that of a three-pulse M-3 connection. The firing angle is less than 30°. Therefore, each device conducts for an angle of 120°.
Vo = 3 x [1/2π ∫ Vmp sinωt d(ωt)] Where, the integration runs from α+π/6 to α+5π/6.
Vo = (3√3/2π) x Vmp x cosα.

43)Find the expression for average output voltage for the given circuit if firing angle is greater than 30°. Take Vmp = secondary side maximum value of phase voltage.



a) (3√3Vmp/2π) x cosα
b) (3√3Vmp/2π) x (1+cosα)
c) (3√3Vmp/2π) x [1+cos(30+α)].
d) (3√3Vmp/2π) x [3+cos(30+α)].

Vo = 3 x [1/2π ∫ Vmp sinωt d(ωt)] Where, the integration runs from α+π/6 to π. Because conduction takes place from 30 + α to 180° for T1 and than the waveform is symmetrical for all other SCRs.
Vo = (3√3/2π) x Vmp x [1+cos(30+α)].

44) A three-phase half-wave controlled converter is fed from a 3-phase, 400 V source and is connected to a load which takes a constant current of 36 A. Find,the value of average output voltage and average current rating of SCR for a firing angle of 30°.
a) 234 V, 36 A
b) 234 V, 12 A
c) 135 V, 36 A
d) 135 V, 12 A

Vo = (3√3/2π) x Vmp x cosα = 233.874 V.
Ia = Io/3 = 12 A.

45) A three phase full converter will require _____ number of SCRs.
a) 3
b) 6
c) 9
d) 2

46) A three phase six pulse full converter works as a ac to dc converter for firing angles in the range
a) α > 90
b) 90 < α < 180
c) 0 < α < 90
d) 0 < α < 360

47) For a three phase full controlled converter, with 3 thyristors in the upper or positive group and 3 thyristors in the lower or negative group, at any given time
a) two thyristors are conducting from each group
b) one thyristor is conducting from each group
c) one thyristor is conducting from either of the groups
d) all 6 thyristors are conducting at a time

48) For a three-phase full controlled converter with R load, the average value of output voltage is zero for
a) $\alpha = 0°$
b) $\alpha = 90°$
c) $\alpha = 180°$
d) It can never be zero

For $\alpha = 90$ degrees, the voltage waveform is equally symmetrical about the ωt axis, hence the average value is zero. This can also be found by using the formula for average output voltage, $Vo = (3Vml/\pi) \cos \alpha$,
For $\alpha = 90°$, $\cos\alpha = 0$, Vo = zero.

49) The effect of source inductance on the performance of a 3-phase controlled converter is to
a) increase the average load voltage
b) reduce the average load voltage
c) make the load current continuous
d) remove ripples from the load current

50) A 3-phase full converter has an average output voltage of 365 V for zero degree firing and resistive load. For a firing angle of 90 degree, the output voltage would be
a) 125 V
b) 569 V
c) 365 V
d) zero

51) Which device can be used in a chopper circuit?
a) BJT
b) MOSFET
c) GTO
d) All of the mentioned

52) What is the duty cycle of a chopper ?
a) Ton/Toff
b) Ton/T
c) T/Ton
d) Toff x Ton

53) The load voltage of a chopper can be controlled by varying the

a) duty cycle

b) firing angle

c) reactor position

d) extinction angle

54) Find the output voltage expression for a step down chopper with Vs as the input voltage and α as the duty cycle.

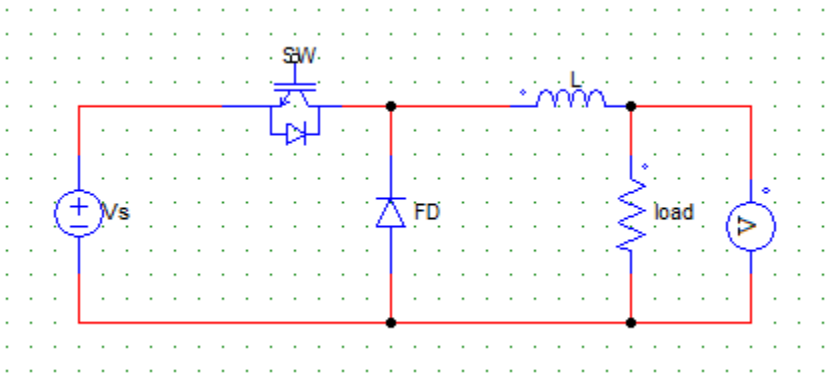
a) Vo = Vs/α

b) Vo = Vs x α

c) Vo = Vs^2/α

d) Vo = 2Vs/απ

55) The below given figure is that of a _____ (IGBT is used as a chopper switch)



a) step-up/step-down chopper

b) step-down chopper

c) step-up chopper

d) none of the mentioned

56) In a step down chopper, if Vs = 100 V and the chopper is operated at a duty cycle of 75 %. Find the output voltage.

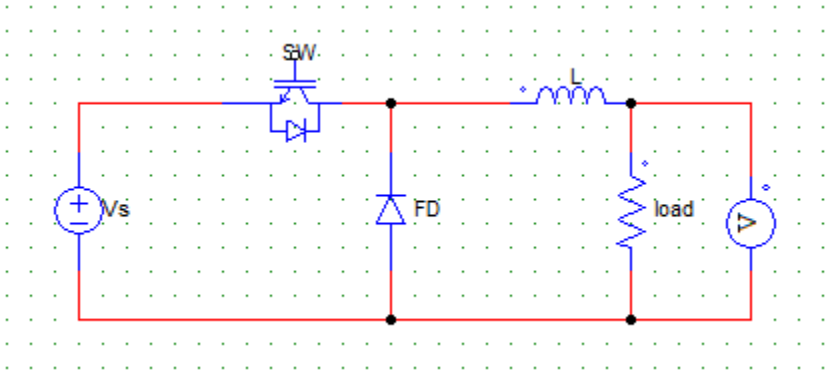
a) 100 V

b) 75 V

c) 25 V

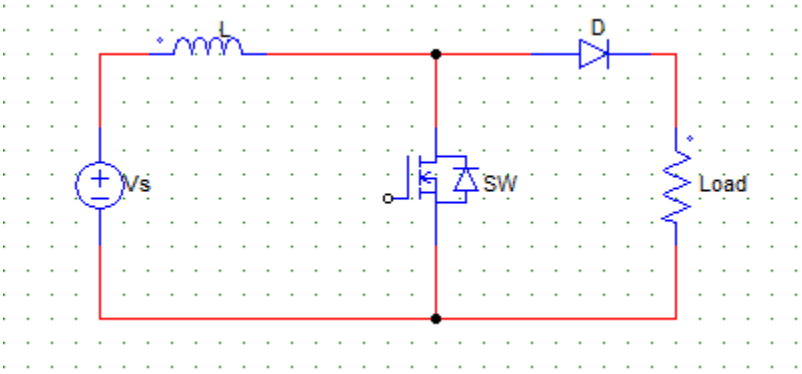
d) none of the mentioned

57) In the below given circuit, when switch (SW) is off



a) voltage is non-zero and current is rising
b) voltage is non-zero and current is decaying
c) voltage is zero and current is rising
d) voltage is zero and current is decaying

58) If Vo is the output voltage and Vs is the input DC voltage, then for the below given circuit.

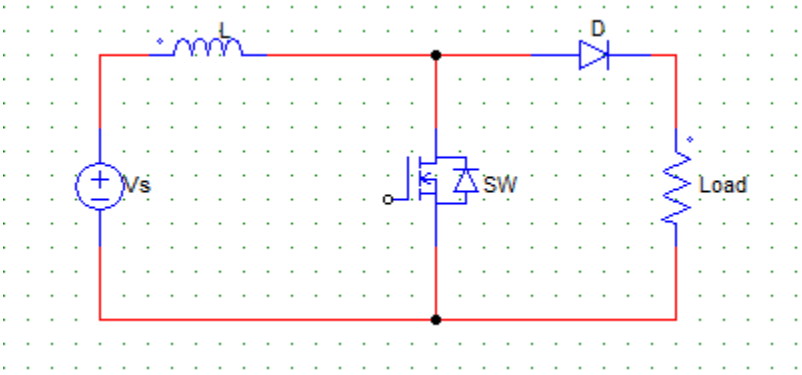


a) Vo = Vs
b) Vo < Vs
c) Vo > Vs
d) Insufficient information

59) Find the expression for output voltage for a step-up chopper, assume linear variation of load current and α as the duty cycle.

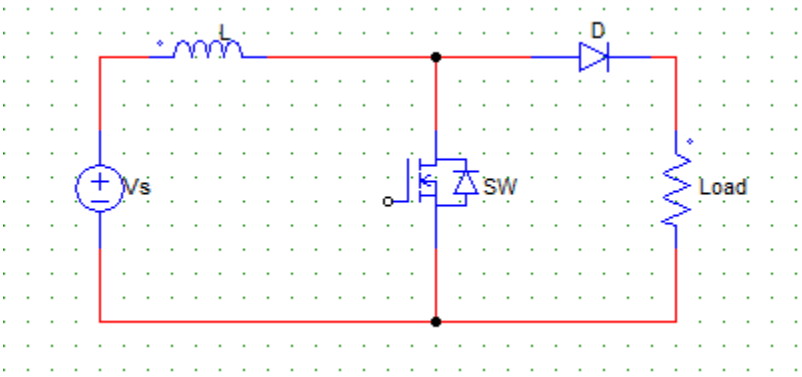
a) Vs
b) Vs/α
c) Vs/(1-α)
d) Vs/√2

60) In the below given circuit, when switch (SW) is first switched closed and then opened the



a) voltage is non-zero and current is rising
b) voltage is zero and current is decaying
c) voltage is non-zero and current is decaying
d) voltage is zero and current is rising

61) What is the voltage across the load when the SW is first closed and then opened?

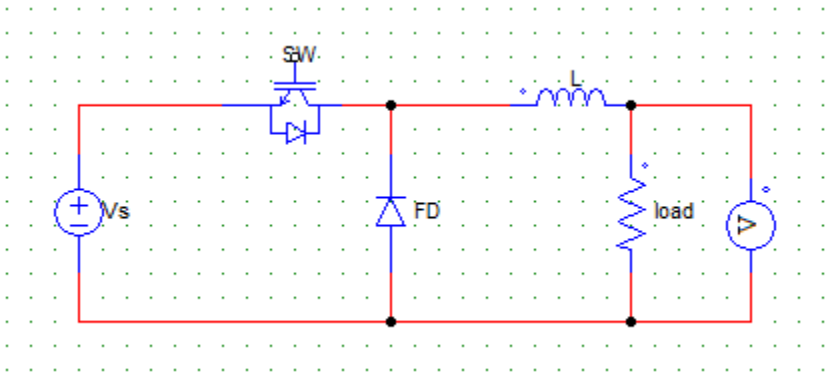


a) Vs
b) duty cycle x Vs
c) Vs + L (di/dt)
d) zero

62) For a step-down chopper, find the rms value of output voltage. Let α be the duty cycle and Vs be the input voltage.

a) α x Vs
b) Vs/α
c) √α x Vs
d) Vs/2

63) For the below given circuit, find the output current at the instant of commutation.



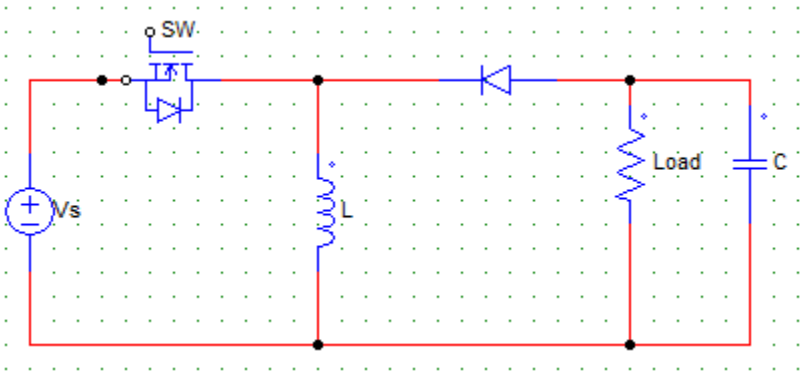
a) Vo/R
b) Vs/R
c) 0
d) α/2

64) A step-up chopper has input voltage of 220 V and output voltage of 660 V. If the conducting time of the IGBT based chopper is 100 µs, compute Toff width of the output voltage pulse.

a) 100 µs
b) 150 µs
c) 50 µs
d) Insufficient data

660 = 220/1-α
α = 2/3 = Ton/T
Ton = 2T/3 = 100µs. This gives chopping period T = 100 x 3/2 = 150 µs
Toff = T – Ton = 150 – 100 = 50 µs.

65) The below given chopper circuit is that of a



a) step-up chopper
b) step-down chopper
c) step-up/step-down chopper
d) none of the mentioned

66) A step down chopper has Vs = 230 V and R = 10 Ω. For a duty cycle of 0.4, the power taken by the chopper is 2097 Watts. Find the chopper efficiency. Take the voltage drop across the chopper switch as 2 V.

a) 98 %
b) 89.96 %
c) 99.14 %
d) 96.54 %

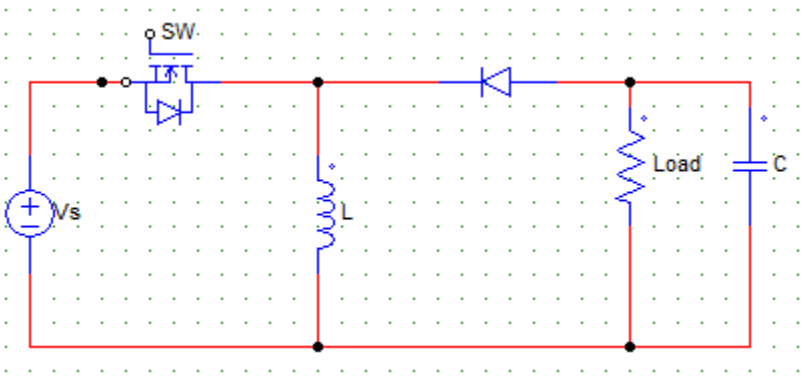
Vo (rms) = √0.4 x (230 – 2) = 144.2 V
Po = $144.2^2/10$ = 2079.3 Watts
Pi = 2097 Watts
η = Po/Pi = 99.14 %.

67) A step down chopper has input dc voltage of 220 V and R = 10 Ω in series with L = 65 mH. If the load current varies linearly between 11 A and 17 A, then find the duty cycle α.

a) 1
b) 0.4
c) 0.6
d) 0.7

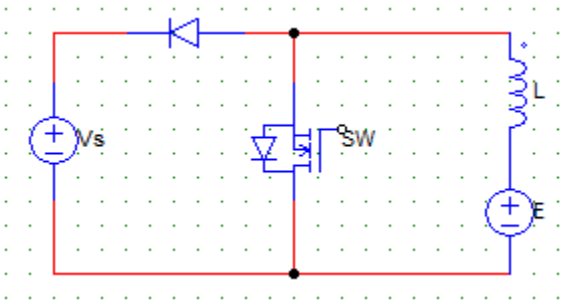
Average load current = 11+17 / 2 = 14A
Average load voltage = IxR = 14×10 = 140V
But, Vo = α x Vs
Therefore, α x 220 = 140, α = 0.7.

68) In the below shown step-up/step-down chopper circuit, when the switch (SW) is on



a) current through inductor rises and the load voltage is zero
b) current through inductor rises and the load voltage is Vs
c) current through inductor decays and the load voltage is zero
d) current through inductor decays and the load voltage is Vs

69) For the type-B chopper shown below



a) during T_{on}, Vo = 0 and L stores energy
b) during T_{on}, Vo is not 0 and L stores energy
c) during T_{on}, Vo = 0 and L releases energy
d) during T_{on}, Vo is not 0 and L releases energy

70) In case of TRC (Time Ratio Control), _____ is varied

a) duty cycle
b) firing angle
c) supply frequency
d) supply voltage magnitude

71) In constant frequency TRC or pulse width modulation scheme, _____ is varied.

a) Vs
b) Ton
c) T
d) f

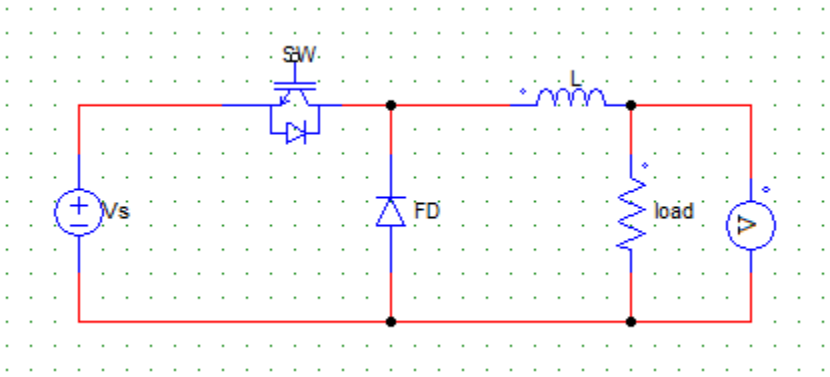
72) In case of variable frequency system _____ is varied

a) T
b) Ton
c) Toff
d) supply frequency

73) In case of frequency modulation system, _____ is kept constant.

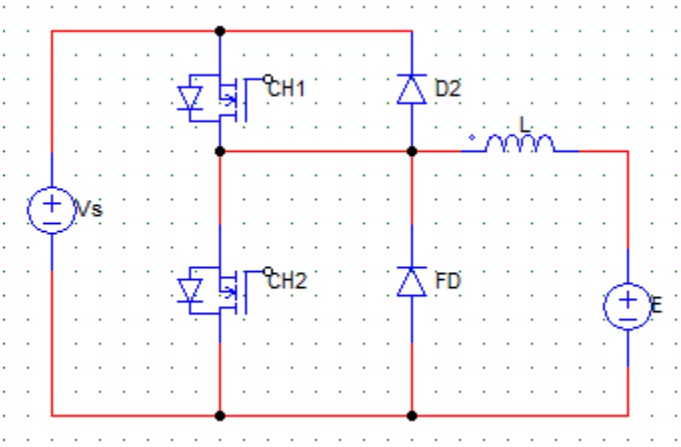
a) T
b) Ton
c) Toff
d) Either Ton or Toff

74) For the step-down chopper shown in the figure below, dc source voltage = 230 V and the voltage drop across the chopper = 2 V. Find the value of rms output voltage for a duty cycle of 0.4.



a) 91.2 V
b) 230 V
c) 144.2 V
d) 145 V

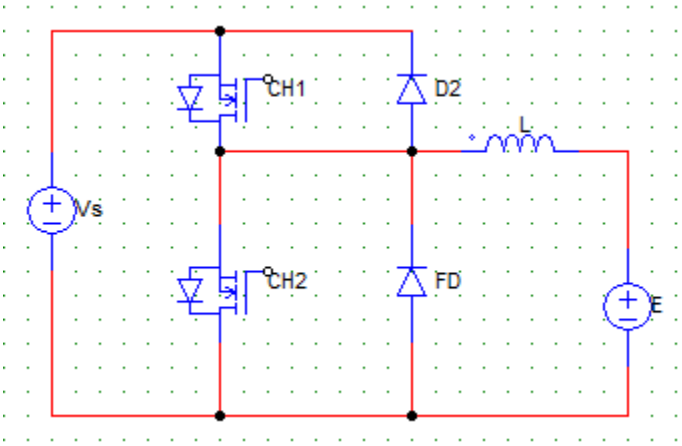
75) In the below given circuit



a) current can never be zero
b) current can flow in both the directions of the load
c) voltage can never be zero
d) none of the mentioned

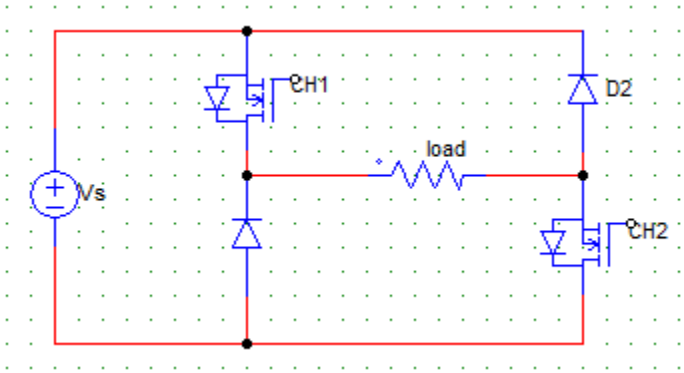
76) Type C chopper consists of
a) two diodes and two switches
b) one diode and one switch
c) one diode and three switches
d) three diodes and two switches

77) In the below given type C chopper circuit, _____ and _____ together operate like type A chopper.



a) CH1, FD
b) CH2, FD
c) CH1, D2
d) CH2, D2

78) What is the load voltage value when both the chopper switches are on?

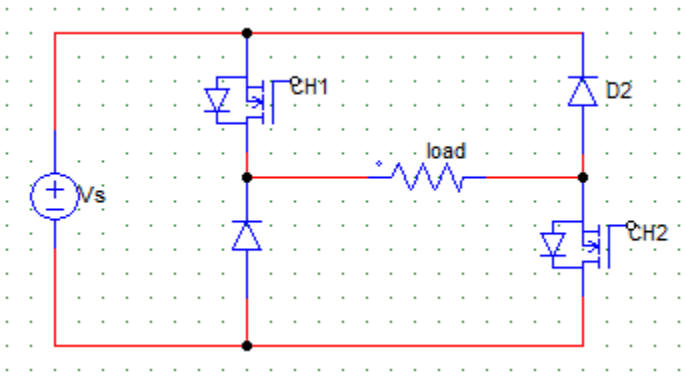


a) 0
b) Vs
c) 2Vs
d) Vs/2

79) For a type D chopper, the average value of output voltage will be positive when

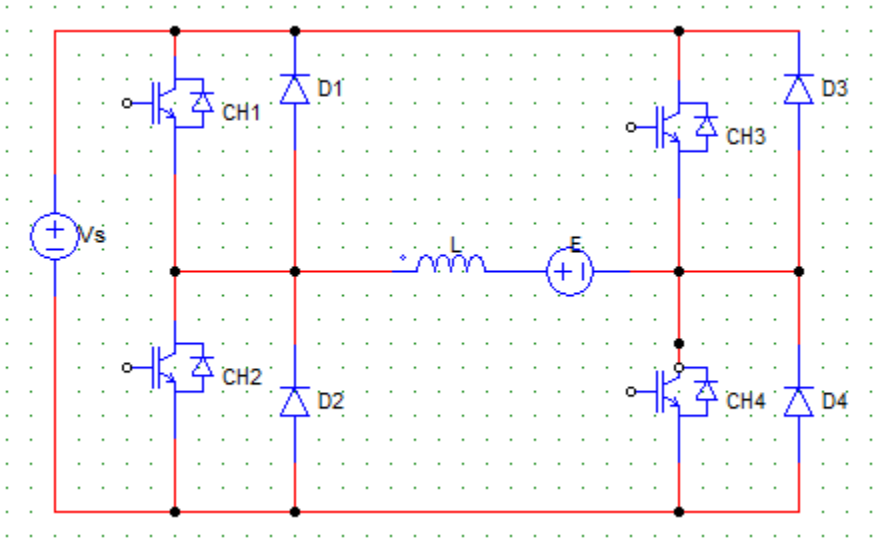
a) $T_{on} = T_{off}$
b) $T_{on} < T_{off}$
c) $T_{off} = 0$
d) $T_{on} > T_{off}$

80) In the below given circuit, if Vo = -Vs then



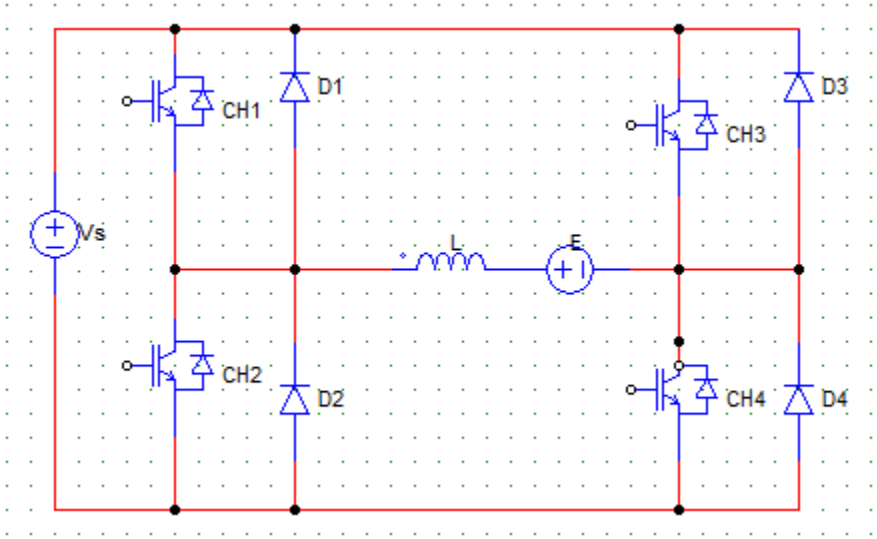
a) both the chopper switches are on
b) only SW2 and D1 is active
c) both the chopper switches are off and both the diodes D1 and D2 are active
d) both the chopper switches are on and both the diodes D1 and D2 are reversed biased

81) Identify the given circuit



a) type D chopper
b) two quadrant type C chopper
c) type E chopper
d) type F chopper

82) For the second quadrant operation of type E chopper



a) Only CH2 is on
b) CH1 and CH2 is on
c) CH2 and CH3 is on
d) CH2 and CH4 is on

83) A step down chopper has Vs = 220 V and is connected to RLE load. With R = 1 Ω, E = 24 V and L = 5 mH. The chopping period is 2000 µs and the on-period is 600 µs. Find the value of minimum steady state output current.

a) 33 A
b) 42 A
c) 0 A
d) 51 A

$I(min) = (Vs/R) \times [(1 - e^{-m})/(1 - e^{-n})] - (E/R)$
$m = Ton / (L/R) = R.Ton/L = 0.12$
$n = T/(L/R) = R.T/L$
$T = 2000\mu s$
$n = 0.4$
$I (min) = (220/1) \times [(1 - e^{-0.12})/(1 - e^{-0.4})] - (24/1) = 51.46$ A.

84) A type A step down chopper has Vs = 220 V and is connected to RLE load. With R = 1 Ω, E = 24 V and L large enough to maintain continuous conduction. Find the average value of load current for a duty cycle of 30 %.

a) 100 A

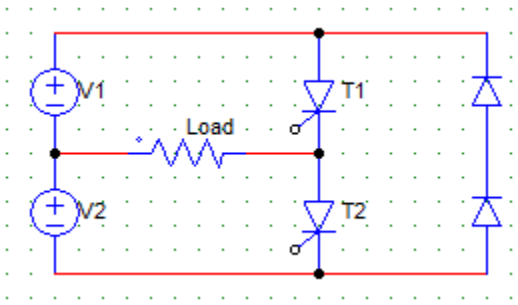
b) 22 A

c) 42 A

d) 16.5 A

I = (αVs – E)/R = 42 A.

85) Identify the circuit given below.



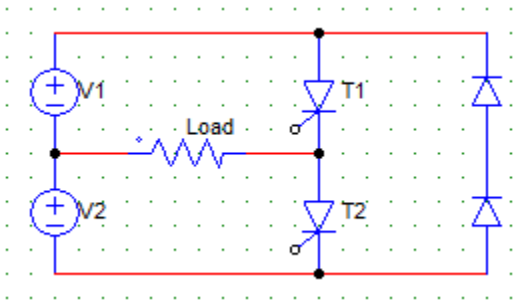
a) Half wave series inverter

b) Full wave series inverter

c) Half wave bridge inverter

d) Half wave parallel inverter

86) What is the voltage across the R load when only T1 is conducting?



a) Vs

b) Vs/2

c) 2Vs

d) zero

87) Single-phase full bridge inverters requires

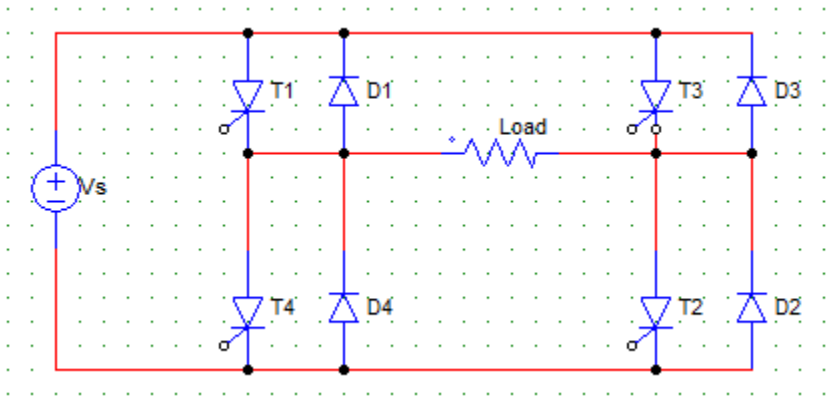
a) 4 SCRs and 2 diodes

b) 4 SCRs and 4 diodes

c) 2 SCRs and 4 diodes

d) 2 SCRs and 2 diodes

88) Identify the circuit given below.



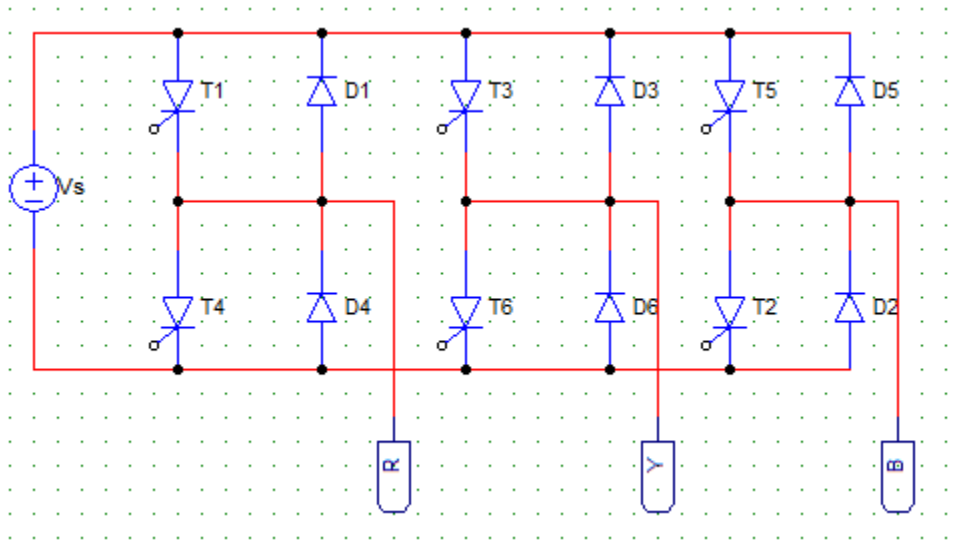
a) Full wave series inverter
b) Half wave series inverter
c) Full wave bridge inverter
d) Full wave parallel inverter

89) For a full wave bridge inverter, the output voltage (Vo)
a) Vo = Vs/2 for 0 < t < T/2
b) Vo = Vs for 0 < t <T/2
c) Vo = Vs for T/2< t < T
d) Vo = -Vs for T/2< t < 3T/2

90) For a half wave bridge inverter, the output voltage
a) Vo = – Vs/2 for 0 < t < T/2
b) Vo = – Vs/2 for T/2< t<T
c) Vo = – Vs for 0 < t < T/2
d) Vo = Vs/2 for T/2< t < T

91) The Total Harmonic Distortion (THD) is the ratio of
a) rms value of all the harmonic components to the rms value of the fundamental component
b) average value of all the harmonic components to the rms value of the fundamental component
c) rms value of all the third harmonic component to the rms value of the fundamental component
d) rms value of all the fundamental component to the rms value of all the harmonic components

92) The below given inverter circuit is a __ step inverter.

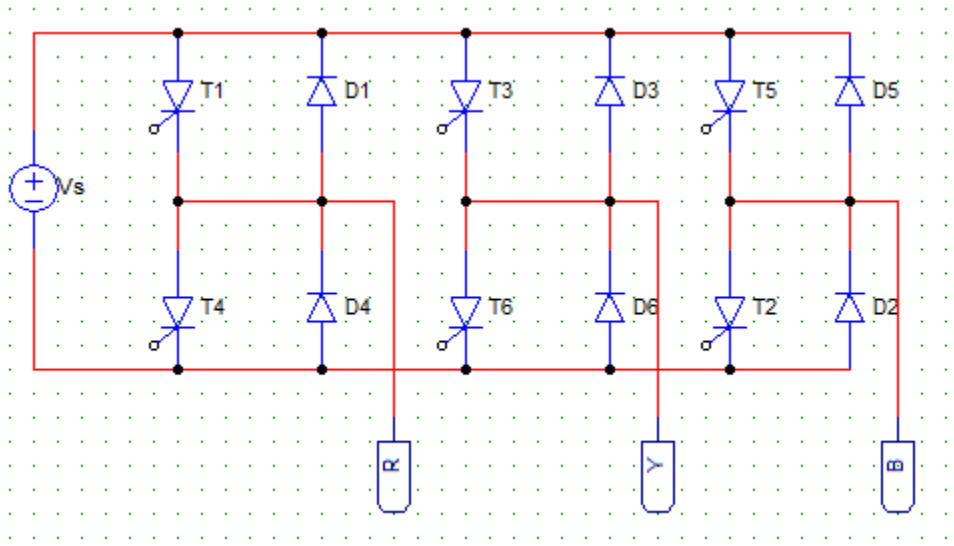


a) 3
b) 2
c) 6
d) none of the mentioned

93) In the 180° mode VSI, _____ devices conduct at a time.
a) 5
b) 2
c) 3
d) 4

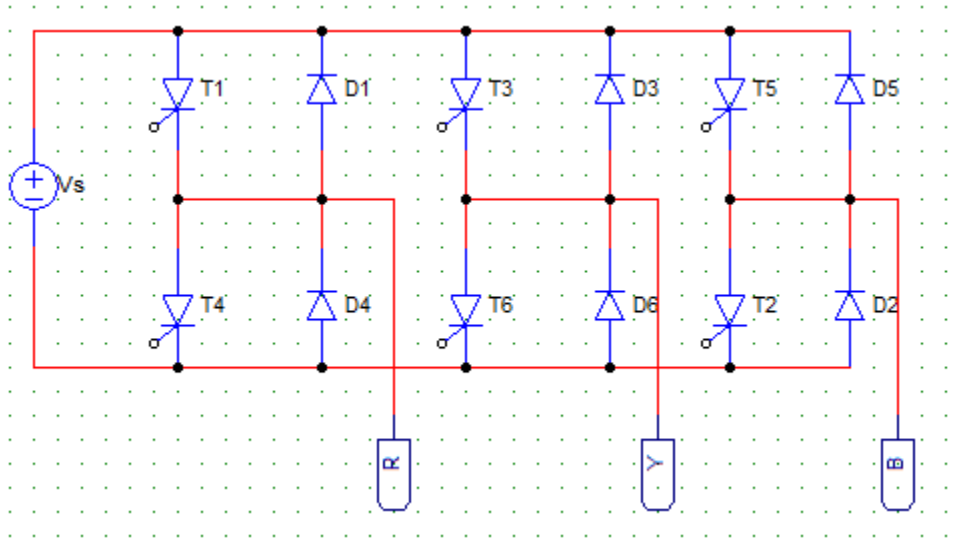
94) What is the peak value of phase voltage in case of 3-phase VSI with 180° mode. The supply side consists of a constant dc voltage source of Vs.
a) Vs
b) 3Vs/2
c) 2Vs/3
d) 3Vs

95) The conducting SCRs from 180 to 240 degrees would be



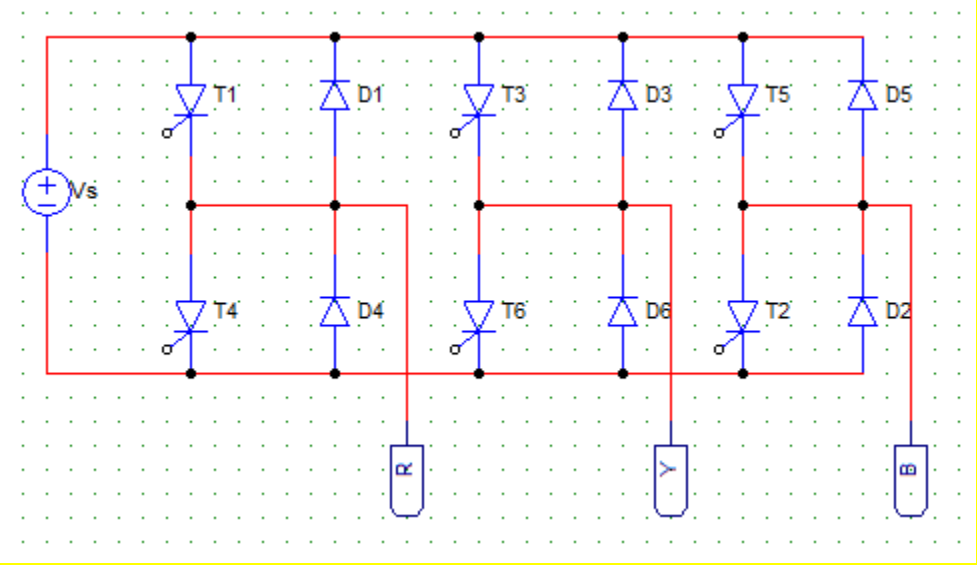
a) 6, 1, 2
b) 2, 3, 4
c) 3, 4, 5
d) 5, 6, 1

96) What is the R phase voltage when T1, T6 and T5 are conducting? Consider a star connected R load.



a) Vs
b) 2Vs/3
c) Vs/3
d) –Vs/3

97)What are the phase voltages when T6, T1 and T2 are conducting? Consider a star connected R load.



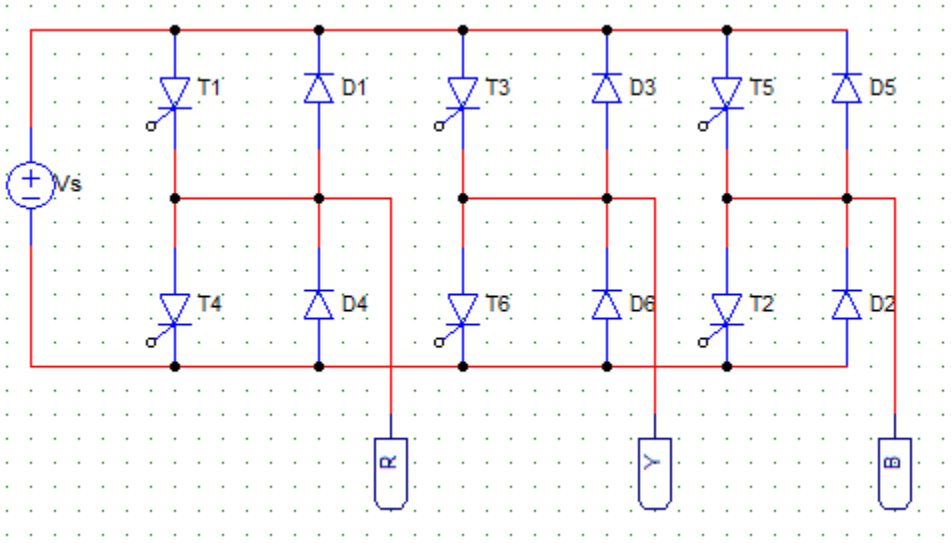
a) $V_{YN} = V_{BN} = Vs/3$, $V_{RN} = 0$
b) $V_{YN} = V_{RN} = – Vs/3$, $V_{BN} = 2Vs/3$
c) $V_{RN} = V_{BN} = – Vs/3$, $V_{YN} = 2Vs/3$
d) $V_{YN} = V_{BN} = – Vs/3$, $V_{RN} = 2Vs/3$

98) What is the maximum line voltage value in case of a three-phase VSI in 180° mode?
a) 2Vs
b) Vs
c) 3Vs
d) 2Vs/3

99) In case of the 120° mode of operation, _____ devices conduct at a time.
a) 2
b) 3
c) 4
d) none of the mentioned

100) Safe commutation can be achieved in case of the _____ operating mode.
a) 180°
b) 120°
c) 360°
d) none of the mentioned

101) What is the R phase voltage and Y phase voltage when only T3 and T4 are conducting? Consider a star connected balanced R load.



a) Vs, -Vs
b) Vs/2, -Vs/2
c) –Vs/2, Vs/2
d) –Vs, Vs

102) In sinusoidal pulse width modulation, _____ wave is compared with a _____ type of wave.

a) square, sinusoidal
b) sinusoidal, triangular
c) sinusoidal, quasi-square
d) none of the mentioned

103) In the sinusoidal pulse width modulation, _____ is the carrier wave signal.

a) square wave
b) triangular wave
c) sinusoidal wave
d) quasi-square wave

104) In the sinusoidal pulse width modulation, _____ is the reference wave signal.

a) square wave
b) triangular wave
c) sinusoidal wave
d) quasi-square wave

105)In sinusoidal pulse width modulation, the comparator output is high when the
a) triangular wave has magnitude higher than the sinusoidal wave
b) sinusoidal wave has magnitude higher than the triangular wave
c) triangular wave has magnitude equal to the sinusoidal wave
d) none of the mentioned

106) The modulation index (MI) is given by
Vr = peak value of the reference wave.
Vc = peak value of the carrier wave.
a) Vr/Vc
b) Vc/Vr
c) (1 + Vc/Vr)
d) 1/(Vc Vr)

107) By controlling the modulation index (MI), _____ can be controlled.
a) gain
b) output frequency
c) harmonic content of the output voltage
d) cosine component of the output voltage

108) In pulse width modulated inverters, the output voltage is controlled by controlling the
a) input frequency
b) modulating index
c) amplification factor
d) none of the mentioned

109) In case of sinusoidal pulse width modulation with MI < 1, the order of the dominate harmonic can be raised by
a) increasing the number of pulses
b) reducing the number of pulses
c) lowering the input voltage frequency
d) raising the input voltage frequency

110) Increasing the number of pulses (N), _____
a) reduces the output voltage amplitude
b) reduces the inverter efficiency
c) improves the inverter efficiency
d) none of the mentioned

111) AC voltage controllers convert
a) fixed ac to fixed dc
b) variable ac to variable dc
c) fixed ac to variable ac
d) variable ac to fixed ac

112) In AC voltage controllers the

a) variable ac with fixed frequency is obtained

b) variable ac with variable frequency is obtained

c) variable dc with fixed frequency is obtained

d) variable dc with variable frequency is obtained

113) In the principle of phase control

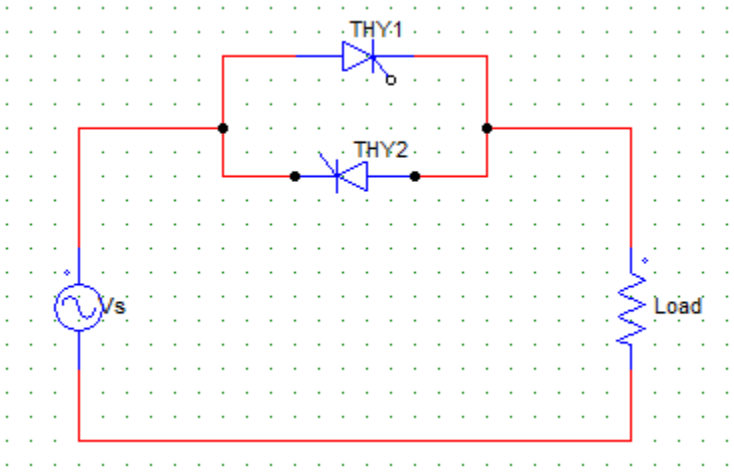
a) the load is on for some cycles and off for some cycles

b) control is achieved by adjusting the firing angle of the devices

c) control is achieved by adjusting the number of on off cycles

d) control cannot be achieved

114) The below given controller circuit is a



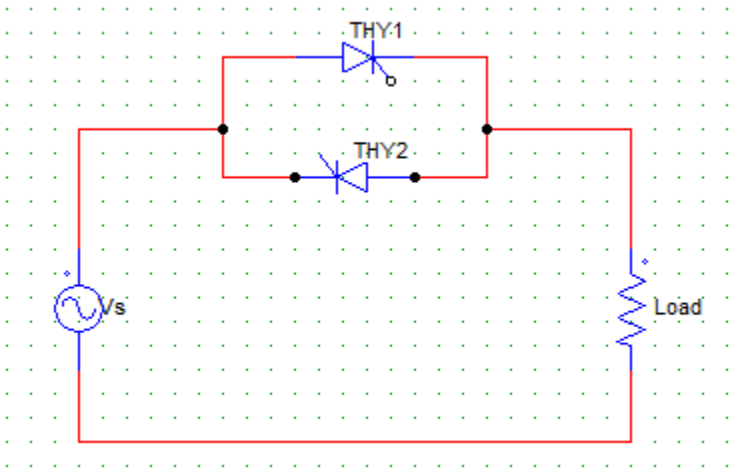
a) half wave controller

b) full wave controller

c) none of the mentioned

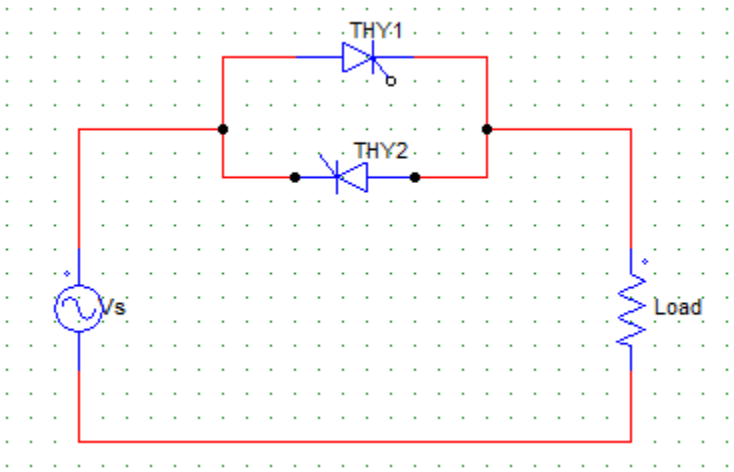
d) will depend upon the firing angle

115) In the below given voltage controller circuit



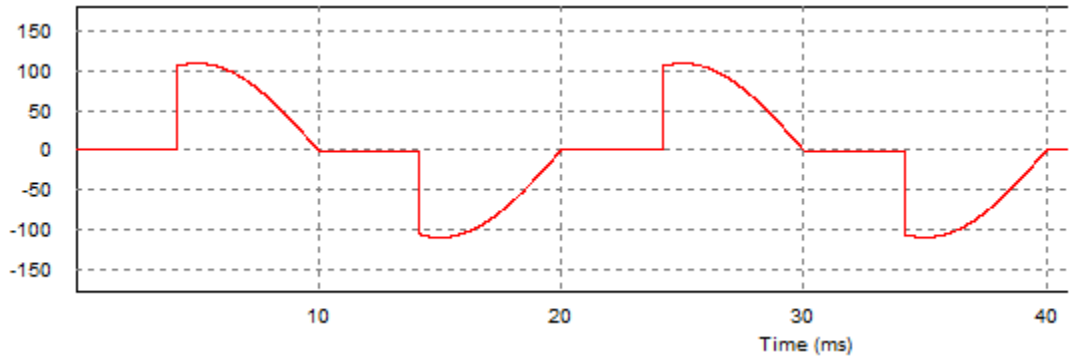
a) only the negative cycle can be controlled
b) only the positive cycle can be controlled
c) both the cycles can be controlled
d) none of the mentioned

116) In the below shown AC converter circuit with firing angle = α for both the devices, T2 will conduct from



a) α to π
b) π + α to 2π
c) π to 2π
d) α to 2π

117) The below given output voltage waveform can be obtained by a



a) half wave ac voltage controller
b) full wave ac voltage controller
c) full wave inverter circuit
d) none of the mentioned